INCH-POUND
MIL-PRF-53064B(AT)
6 February 1996
SUPERSEDING
MIL-H-53064A(ME)
12 February 1992

PERFORMANCE SPECIFICATION

HOSE ASSEMBLY, ELASTOMER: LIGHTWEIGHT, COLLAPSIBLE; LIQUID FUEL

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a lightweight, collapsible, elastomer hose assembly for use with liquid fuels (see 6.1).

1.2 Classification. Hose assemblies will be of the following sizes and classes, as specified (see 6.2):

Sizes, listed in inside diameters (ID):

2-inches (in.)
3-inches (in.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 4720

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Classes:

Class 1 - Coupling Halves, Quick-Disconnect, Cam-Locking Type, Hose Shank; Female Both Ends.
Class 2 - Coupling Halves, Quick-Disconnect, Cam-Locking Type, Hose Shank; Male Both Ends.
Class 3 - Coupling Halves, Quick-Disconnect, Cam-Locking Type, Hose Shank; Female One End, Male Other End.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for addition information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-G-3056 - Gasoline, Automotive, Combat.
MIL-T-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-C-27487 - Coupling Halves, Quick-Disconnect, Cam-Locking Type.

FEDERAL

A-A-52554 - Clamps, Hose, Low Pressure.
A-A-52557 - Fuel Oil, Diesel.

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STANDARDS

FEDERAL

FED-STD-595 - Colors Used in Government Procurement.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

Title 10, CFR, Part 40, and 2 - Radioactive Material.

(Copies of CFR publications may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 29492.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR QUALITY CONTROL (ASQC)

ASQC Z1.4 - Sampling Procedures and Tables for Inspection by Attributes (DoD Adopted).

(Copies of ASQC Standards may be obtained from The American Society for Quality Control, P.O. Box 305, 611 East Wisconsin Avenue, Milwaukee, WI 53201-4606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D380 - Rubber Hose.
D381 Existent Gum in Fuels by jet Evaporation.
D412 - Rubber Properties in Tension.
D413 - Rubber Property - Adhesion to Flexible Substrate.

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Nominal Fine Torque

D471	- Rubber Property - Effect of Liquids.
D518	- Rubber Deterioration - Surface Cracking.
D910	- Aviation Gasolines.
D1149	- Rubber Deterioration - Surface Ozone Cracking in a Chamber (Flat Specimen).

(Copies of ASTM publications may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2). The hose assembly shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Unless otherwise specified herein, materials used shall be selected by the manufacturer. The hose assemblies shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide against corrosion and deterioration during storage and operational conditions experienced. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Recovered materials shall be used to the maximum extent practicable (see 4.6.1). Asbestos, cadmium, and radioactive material will not be used in this item. Radioactive material is defined by 1 (Title 10, Code of Federal Regulations, Part 40, and 2). Other radioactive material in which the radioactivity is greater than 0.002 microcuries per gram or 0.01 microcuries total activity for the item.

3.3 Design and construction. The hose shall consist of a compounded elastomer inner tube, one or more plies of synthetic fiber reinforcement material, static wire, and compounded elastomer cover (see 4.6.2). The hose shall not leak. The color of the hose shall be sand matte in accordance with FED-STD-595, color chip 33446. The tube shall be fabricated of an elastomer compounded material which is resistant to hydrocarbon fuels, and shall not have any deleterious effects upon any type and grade of fuel conforming to MIL-G-3056, ASTM D910, MIL-T-5624, or A-A-52557. The tube shall be smooth and free of pits. The cover shall be of an elastomer compounded for fuel, oil, sunlight, water and ozone resistance. The cover shall be uniform in thickness.

3.3.1 Static wire. The static wires shall be embedded within or between the plies to form an angle of not less than 45 degrees (°) to the longitudinal axis of the hose or woven along with

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the longitudinal yarn. The wires shall not break when the hose is coiled, or subjected to internal or external pressure. The static wire shall be either stainless steel or coated copper. Not less than 1.0 in. nor greater than 1.5 in. of each wire shall be in intimate contact with the couplings such that the resistance between the couplings shall not exceed 1.5 ohms per linear foot of hose (see 4.6.13). The static wire shall not reach beyond the first hose band and in no case shall the wire protrude into the flow path.

3.3.2 Color lay-line. A longitudinal black lay-line shall be marked on the outer cover of the hose. It shall be not less than 0.125 in. wide and shall be continuous for the full length of the hose. The lay-line shall remain clear, distinct and legible for the life of the hose. Handling, bending, fuel, or environmental effects shall not adversely affect the lay-line (see 4.6.2).

3.3.3 Reinforcement. The reinforcement of the hose shall be of a carcass of synthetic filament warp yarns interwoven with fillers of synthetic filament yarns or shall be of one or more plies of woven, braided or spiraled synthetic fiber cord. The reinforcement material shall be free of dirt, knots, lumps, or irregularities of twist. The reinforcement shall be evenly applied and thoroughly impregnated with elastomer so that the plies are bonded to their adjacent layers, tube, and cover, so as to produce a unified hose wall.

3.3.4 Couplings and fitting.

3.3.4.1 Coupling halves and coupling caps. Coupling halves (cam-locking type), shall be in accordance with MIL-C-27487, type II or VI, class 1 or 2, as applicable (see 1.2). Type IX dust caps or the type X dust plug conforming to MIL-C-27487 shall be provided for the coupling halves by means of chain attachments or other similar attachments (see 4.6.2).

3.3.4.2 Band fittings. Each coupling shall be banded to the hose by bands conforming to A-A-52554. There shall be 2 bands for each end of every hose. Each band shall be 0.625 in. The bands shall be covered with a non-vivid and non-reflective neoprene sleeve to protect the user (see 4.6.2).

3.4 Performance

3.4.1 Diameter, weight, and length. The length of hose assembly shall be as specified (see 6.2) with a tolerance of ±0.5 in. Hose inside diameter and weight shall be as specified in table I (see 4.6.2).

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TABLE I. Diameters and weight.

Hose diameter (Inner diameter) +0.10-in. - 0.03-in.	Weight per linear foot (Without couplings) ±0.10 pound (lb)
2 in.	0.60 lb (max)
3 in.	0.95 lb (max)

3.4.2 Hydrostatic pressure. The hose and hose assembly shall be tested in conformance to ASTM D380 per elongation, twist, and proof pressure, using water as the test fluid (see 4.6.3). The hose and hose assembly shall be tested at the proof pressure listed in table II. While under pressure, the hose assembly shall not leak, and there shall be no indication of imperfections, slippage or pull-out of the hose from the coupling. The length of the hose shall not change more than ±2 percent (%).

3.4.3 Minimum burst. The hose and hose assembly shall be tested for burst pressure. When tested, each hose size shall have a minimum burst pressure as specified in table II (see 4.6.4).

TABLE II. Pressure for hose assembly.

Size I.D. (in.)	Operating pressure (psi)	Proof pressure (psi) minimum	Burst pressure (psi) minimum
2	75	200	300
3	75	200	300

3.4.4 Tensile strength. The tensile strength of the tube and cover shall not be less than 1250 pounds per square inch (psi) and 1500 psi respectively (see 4.6.5).

3.4.5 Tensile strength after immersion. The tensile strength of the tube and cover after immersion shall not be less than 600 psi (see 4.6.6).

3.4.6 Ultimate elongation. The ultimate elongation of the tube and cover shall be not less than 300 %, and the elongation after immersion shall be not less than 200 % (see 4.6.7).

3.4.7 Original adhesion and adhesion after immersion. The adhesion between the tube and plies, between the plies, and between the cover and plies before and after immersion shall not be less than 10 pounds force (lbf) (see 4.6.8) and 6 lbf respectively (see 4.6.9).

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3.4.8 Low temperature flexibility. Low temperature flexibility shall be determined at a minimum pressure of 200 psi, there shall be no evidence of cracking, or lack of flexibility, and the hose shall not leak (see 4.6.10).

3.4.9 Ozone resistance. The cover material shall be viewed under 7-power magnification, the cover shall show no evidence of cracking, checking or other irregularities at the end of the exposure time (see 4.6.11).

3.4.10 Existent and heptane-washed gum content. The hose shall be tested for gum content, the existent gum content shall not be more than 0.0175 pounds per gallon (lb/gal) of test fluid. The heptane-washed gum content shall not be more than 0.00042 lb/gal (see 4.6.12).

3.4.11 Electrical resistance. The electrical resistance of the hose shall be determined from coupling to coupling, and shall not exceed 1.5 ohms per linear foot of hose (see 4.6.13).

3.5 Safety marking. The hose shall be marked a maximum of 15 foot (ft) intervals with the following "CAUTION: DO NOT DRAG OVER IMPROVED SURFACES". The lettering shall be a minimum of 0.50 in. and be black Gothic capitals.

3.6 Workmanship. The hose assembly shall conform to the quality requirements as specified herein. There shall be no cuts, nicks, bruises, blisters, lumps, or sharp edges on the hose assembly. The hose assembly shall be clean and smooth. The hose assembly shall have no other defects affecting appearance or serviceability.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Quality conformance inspection (see 4.3).

4.2 First article inspection. The first article shall be examined as specified in 4.4, in order shown. The first article shall be subjected to the tests of 4.6.3 through 4.6.12 (see table III).

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TABLE III. Hose assembly tests.

Test	Requirements	Test	First article	Quality conformance	Individual
Hydrostatic pressure	3.4.2	4.6.3	X	X	X
Minimum burst	3.4.3	4.6.4	X	X	
Tensile strength	3.4.4	4.6.5	X	X	
Tensile strength after immersion	3.4.5	4.6.6	X	X	
Ultimate elongation	3.4.6	4.6.7	X	X	
Ultimate elongation after immersion	3.4.6	4.6.7	X	X	
Original adhesion	3.4.7	4.6.8	X		
Adhesion after immersion	3.4.7	4.6.9	X	X	
Low temperature flexibility	3.4.8	4.6.10	X	X 1/	
Ozone resistance	3.4.9	4.6.11	X	X	X
Existent and heptane-washed gum content	3.4.10	4.6.12	X		
Electrical resistance	3.3.1	4.6.13		X	

1/ One lot testing required for each 10 lots submitted.

4.3 Conformance inspection. Conformance inspection shall consist of the examinations of 4.4 and the tests of 4.5.

4.4 Examinations.

4.4.1 Sampling. Samples from an inspection lot for conformance inspection shall be selected in accordance with ASQC Z1.4. Each sample selected shall be inspected as specified herein for the defects listed in table IV and for the tests marked X in table III test sections cut from the hose shall be of sufficient length for the performance of their required test.

TABLE IV. Examinations.

Category	Defect	Method of examination
Major:		
101	Material not as specified (see 3.2)	Visual
102	Materials not corrosion resistant or treated to be made corrosion resistant for the aapplicable storage and operating environment (see 3.2).	Visual

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TABLE IV. Examinations - Continued.

Category	Defect	Method of examination
103	Used, rebuilt, or remanufactured components, pieces or parts incorporated in the hose assembly (see 3.2).	Visual
104	Hose not as specified (see 3.3).	Visual
105	Color of hose not as specified (see 3.3).	Visual
106	Tube not as specified (3.3).	Visual
107	Hose cover scuffed, or bruised, so that braid is exposed (see 3.3).	Visual
108	Hose cover not clean and smooth (see 3.3).	Visual
109	Hose color lay-line not as specified (see 3.3.2).	Visual
110	Fittings and couplings not as specified (3.3.4.1 and 3.3.4.2).	Visual
111	Dimensions not as specified (see 3.2).	Visual
112	Workmanship not as specified (see 3.6).	Visual
113	Safety marking not as specified (see 3.5).	Visual
114	Electrical resistance not as specified (see 3.4.11).	SIE 1/

1/ SIE = Standard Inspection Equipment.

4.5 Individual tests. Each hose assembly produced shall be subjected to those tests marked "X" in column 6 of table III.

4.6 Method of inspection.

4.6.1 Materials. Conformance to 3.2 and 3.3.1 shall be determined by inspection of contractor records providing proof or certification that materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.6.2 Defects. Conformance to 3.3, 3.3.2, 3.3.4.1, 3.3.4.2, 3.4.1, 3.5, and 3.6 shall be determined by examination for the defects listed in table IV. Examination shall be visual, tactile, or by measurement with SIE.

4.6.3 Hydrostatic pressure. To determine conformance to 3.4.2, the hose assembly shall be tested to a proof pressure of 200 psi. The electric bond shall be determined while pull test pressure is applied. Elongation shall be ±2 %. Twist shall be 20 degrees per foot for 2 in. and 8 degrees per foot for 3 in. hose.

4.6.4 Minimum burst. To determine conformance to 3.4.3. The hose and hose assembly shall be tested in conformance to ASTM D380 for burst pressure. A hose not less than 3 ft nor

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more than 6 ft shall be cut from hose selected, and shall be subject to a minimum burst pressure of 300 psi, using water as the test fluid..

4.6.5 Tensile strength. To determine conformance to 3.4.4, the tensile strength for the tube and cover shall be determined in conformance to ASTM D471, and shall not be less than 1250 psi and 1500 psi respectively.

4.6.6 Tensile strength after immersion. To determine conformance to 3.4.5, the tube and cover shall be tested in conformance to ASTM D471 at a temperature of 73 ±2 degrees Fahrenheit (°F), at an immersion period of 70 hours (h), and the immersion fluid shall be reference fuel D. After immersion, the tensile strength shall not be less then 600 psi.

4.6.7 Ultimate elongation. To determine conformance to 3.4.6, the ultimate elongation of the tube and cover before and after immersion shall not be less than 300 % before, and 200 % after immersion.

4.6.8 Original adhesion. To determine conformance to 3.4.7, the original adhesion shall be tested in conformance with the machine method of ASTM D413, and shall not be less than 10 lbf.

4.6.9 Adhesion after immersion. To determine conformance to 3.4.7, the hose assembly shall be tested in conformance to ASTM D413, for 70 h at 93 ±2 °F in reference fuel C. Within 15 min of removal from the test fluid, the adhesion shall not be less than 6 lbf.

4.6.10 Low temperature flexibility. To determine conformance to 3.4.8, the hose shall be tested in accordance with ASTM D380, at a minimum pressure of 200 psi, at a temperature of -25 ±2 °F.

4.6.11 Ozone resistance. To determine conformance to 3.4.9, the cover shall be prepared in accordance with ASTM D518, procedure A and tested in accordance with ASTM D1149. The cover shall be mounted in a 20 % elongated position, conditioned for 29 h in an ozone-free atmosphere. After conditioning, the cover shall be exposed for 72 h at 104 ±1 °F at an atmosphere containing 50 ± 10 parts per hundred million (pphm) of ozone.

4.6.12 Existent and heptane-washed gum content. To determine conformance to 3.4.10, the hose shall be tested in accordance to ASTM D381. A hose not less than 14 in. long shall be stoppered at one end with a 2 in. plug and secured with a clamp. The hose shall be filled with fuel and similarly secured at the opposite end. The hose shall be tested for a period of seven days by agitating the fuel by means of movement every 24 h period. The agitating process is performed by moving the hose from a vertical to a horizontal position at a rate of 2 cycles per minute for 5 min at each 24 h period. At the end of the seven day storage period, the fuel shall

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be agitated in the same manner as above for 5 min and than immediately removed from the hose. The removed fuel shall be tested in accordance to ASTM D381 (air jet solvent wash method). A control sample of reference fuel D shall also be tested in accordance to ASTM D381. The valve of the existent gum content of the control fuel shall be subtracted from that of the test fuel to give a valve of existent gum. The residue obtained from the existent gum. The residue obtained from the existent gum determination, shall then be used to determine heptane-washed gum in accordance to ASTM D381, procedures 10.8 through 10.12.

4.6.13 Electrical resistance. To determine conformance to 3.4.11, the electrical resistance of each hose shall be determined from coupling to coupling with an ohmmeter, and shall not exceed 1.5 ohms per linear foot of hose.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. This lightweight, collapsible hose is intended for use with forward area refueling systems which discharge liquid fuels such as automotive gasoline, aviation fuel, jet fuel, and diesel fuel.

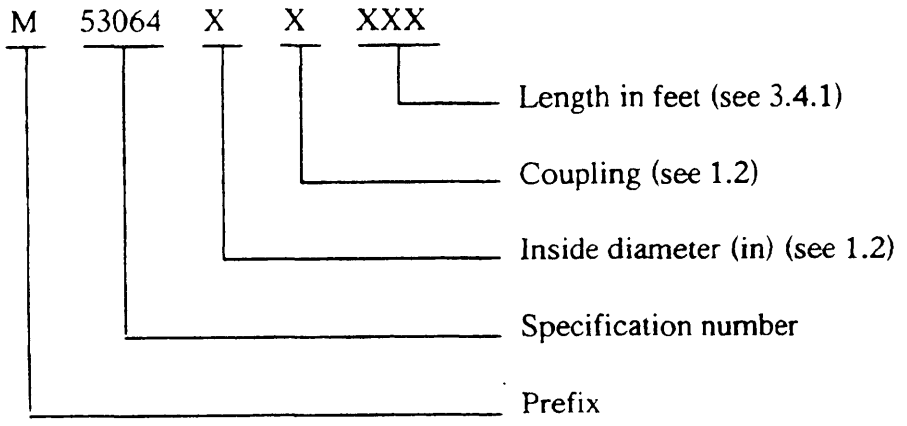
6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Size of hose assembly required (see 1.2).
c. Class of coupling halves (see 1.2).
d. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2 and 2.3).
e. When a first article is not required for inspection (see 3.1).
f. Length of hose required (see 3.4.1).

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g. Selection of applicable level of packaging (see 5.1).
h. PIN number (see 6.3).

6.3 Part or identification system. The part or identification number (PIN) of hose assemblies covered by this specification will be designed in the following form (see 6.2).



6.4 Subject term (key word) listing.

Couplings
Cover
Gasoline
Tube

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - AT

Review activity:
Army - CD
DLA - CS

Preparing activity:
Army - AT

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5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY	
a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial: (810) 574-8745 (2) AUTOVON: 786-8745
c. ADDRESS (Include Zip Code) Commander U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, [illegible] MI [illegible]	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT [illegible] Standardization Officer 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. FSC 4720

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-PRF-53064B(AT)	960131

3. DOCUMENT TITLE
Hose Assembly, Elastomer: Lightweight, Collapsible; Liquid Fuel

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

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